



**Year 11 Mathematics Advanced
Assessment Task 1
Term 2, Week 3, 2024**

Setter: BRA

Name: _____

Teacher: _____

- Answer Questions in the space provided.
- Answer multiple choice questions on the question paper by circling the correct alternative.
- Show all relevant mathematical reasoning and/or calculations.
- Use blue or black PEN only.
- NESA approved calculators are permitted in this task.
- Your own Formula Sheet can be used with this task.
- Marks may be deducted for insufficient or illegible work.
- Extra writing space is provided at the back of this booklet.
If you use this space, clearly indicate which question you are answering.
- Total possible marks: **32**
- Time allowed: **45 minutes plus 2 minutes Reading Time**

Wednesday , 15th May 2024

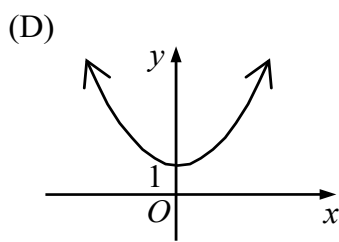
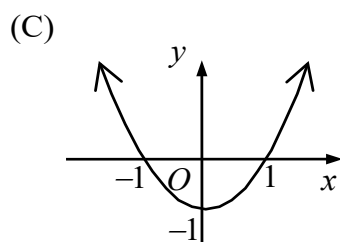
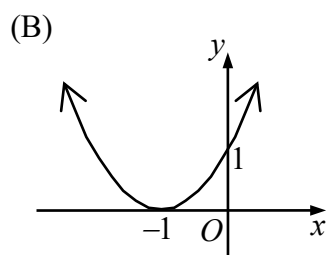
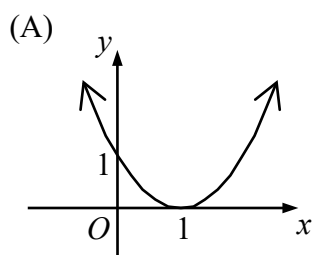
Q1	Q2	Q3	Q4	Q5	Q6	Q7	TOTAL
/6	/6	/4	/4	/4	/4	/4	/32

Question 1 (6 marks)**Marks**

(a) Circle your correct alternative.

1If $ab = 12$, $bc = 20$ and $ac = 15$, then a possible value for abc is:(A) -600 (B) 60 (C) 360 (D) 600

(b) Circle your correct alternative.

1Which graph drawn below best represents $y = (x + 1)^2$?

Question 1 Continued

- (c) Given $M = \frac{m}{\sqrt{1 - \frac{v^2}{c^2}}}$, find M if $m = 550$, $v = 2.75 \times 10^8$ and $c = 3 \times 10^8$.

2

Give your answer correct to five significant figures.

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- (d) Simplify completely $\frac{4a^2b - 8ab^2}{4ab^2 - 8a^2b}$.

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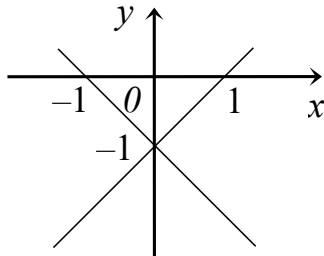
End of Question 1

Question 2 (6 marks)**Marks**

(a) Circle your correct alternative.

1

The equation $x^2 - (y+1)^2 = 0$ represents a relation between x and y in the xy -plane. Its sketch is shown below.



The relation can be classified as:

- (A) One-to-one
- (B) One-to-many
- (C) Many-to-one
- (D) Many-to-many

(b) Circle your correct alternative.

1

Consider the following two statements:

- I. Two irrational zeroes of a quadratic function cannot be equal.
- II. If α and β are distinct zeroes of the quadratic function $f(x) = ax^2 + bx + c$, where a , b and c are real constants, then $a\alpha + b + a\beta = 0$.

Which of the above statements are true?

- (A) Neither I nor II
- (B) I only
- (C) II only
- (D) Both I and II

Question 2 Continued

(c) Solve $\frac{x-5}{4} = \frac{6}{x}$.

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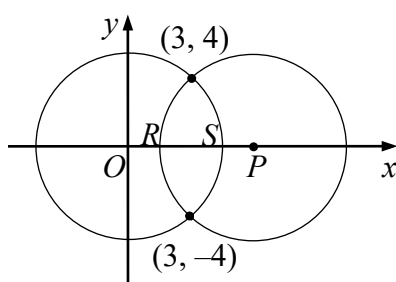
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(d)



Not to Scale

2

In the diagram above, O and P are centres of circles with equal radii, which intersect at $(3, 4)$ and $(3, -4)$. The circles cut the x -axis at R and S as shown.

Find the value of the length RS . Explain your reasoning carefully.

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End of Question 2

Question 3 (4 marks)**Marks**

(a) Sketch a function $f(x)$ that possesses all of the following properties:

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1. $f(x)$ is an odd function.
2. $f(x)$ has horizontal asymptotes $y = \pm 2$.
3. $f(x)$ has vertical asymptotes $x = \pm 2$.
4. $f(x)$ has domain $x < -2$ together with $x > 2$ as well as $x = 0$.

(b) Write $\frac{\sqrt{2}+1}{\sqrt{2}-1}$ in the form of $a + b\sqrt{x}$, where a , b and x are integers.

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End of Question 3

Question 4 (4 marks)**Marks**

A function is defined by the formula $f(x) = \frac{\sqrt{-x}}{\sqrt{x+1}}$.

- (a) Determine the domain of the function.

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- (b) Determine the range of the function.

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- (c) Given f is a one-to-one function, what is the relationship between x_1 and x_2 , if $f(x_1) = f(x_2)$? Explain your reasoning carefully.
You may assume x_1 and x_2 are in the domain of f .

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End of Question 4

Question 5 (4 marks)

- (a) If $f(x) = \sqrt{x}$ sketch the graph of $y = f(x-1)$ showing intercepts with the coordinate axes. **1**

- (b) $P(1, 2)$ is a point on the line $3x + y - 5 = 0$. Find the coordinates of the points on this line, so that their distance from P is $\sqrt{10}$ units. **3**

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End of Question 5

Question 6 (4 marks)

- (a) Simplify completely $\sqrt{54x} \div \sqrt{6}$

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- (b) Factorise fully $a^2 - 2ab + b^2 + 3a - 3b + 2$.

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End of Question 6

Question 7 (4 marks)

- 1**

[illegible]

- 3

For x a positive integer, how many positive integers are there such that both x and $x + 99$ are perfect squares? Explain carefully your reasoning.

Hint: Let $x = m^2$ and $x + 99 = n^2$, where m and n are positive integers.

[illegible]

End of Assessment Task

Extra writing space

If you are using this space indicate clearly which question(s) you are answering & write ‘see extra writing space’ on the original question space provided for that question.

[illegible]

[illegible]



**Year 11 Mathematics Advanced
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/6	/6	/4	/4	/4	/4	/4	/32

Question 1 (6 marks)

Marks

(a) Circle your correct alternative.

1

If $ab = 12$, $bc = 20$ and $ac = 15$, then a possible value for abc is:

(A) - 600

(B) 60

(C) 360

(D) 600

$$(ab)(ac) = 12 \times 15$$

$$\Rightarrow a^2 bc = 180 \quad \& \quad bc = 20$$

$$\Rightarrow 20a^2 = 180$$

$$\Rightarrow a^2 = 9$$

$$\Rightarrow a = \pm 3$$

$$\therefore abc = a \times (bc)$$

$$= \pm 3 \times 20$$

$$= \pm 60$$

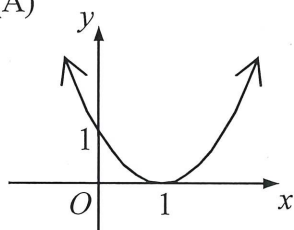
$\therefore (B)$

(b) Circle your correct alternative.

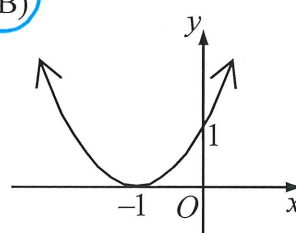
1

Which graph drawn below best represents $y = (x + 1)^2$?

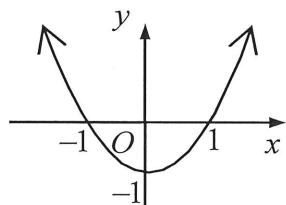
(A)



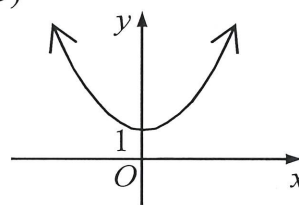
(B)



(C)



(D)



$y = (x+1)^2$ is a translation of $y = x^2$ one unit to the LEFT. $\therefore (B)$.

Question 1 Continued

- (c) Given $M = \frac{m}{\sqrt{1 - \frac{v^2}{c^2}}}$, find M if $m = 550$, $v = 2.75 \times 10^8$ and $c = 3 \times 10^8$.

2

Give your answer correct to five significant figures.

$$M = \frac{550}{\sqrt{1 - \left(\frac{2.75 \times 10^8}{3 \times 10^8}\right)^2}}$$

N.B. $\frac{2.75}{3} = \frac{11}{12}$ & so equivalent calculation is

$$= \frac{550}{\sqrt{1 - \left(\frac{2.75}{3}\right)^2}} \quad \frac{550}{\sqrt{1 - \left(\frac{11}{12}\right)^2}}$$

$$= 1376.195133... \text{ (Calculator)}$$

$$= 1376.2 \text{ (5 s.f.)}$$

Correct bald
answer: 2 MARKS

- (d) Simplify completely $\frac{4a^2b - 8ab^2}{4ab^2 - 8a^2b}$.

$$= \frac{4ab(a - 2b)}{4ab(b - 2a)} \quad \checkmark \text{ 1 mark}$$

$$= \frac{a - 2b}{b - 2a} \quad \checkmark \text{ 1 mark}$$

BALD ANSWER: 1 mark only.

✓ for correct calculation
✓ for correct # of sig. figures. 2

End of Question 1

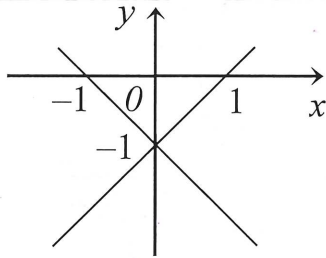
Question 2 (6 marks)

Marks

(a) Circle your correct alternative.

1

The equation $x^2 - (y+1)^2 = 0$ represents a relation between x and y in the xy -plane. Its sketch is shown below.



The relation can be classified as:

- (A) One-to-one
- (B) One-to-many
- (C) Many-to-one
- ☒ (D) Many-to-many

The relation is NOT a function and so (A) & (C) are immediately eliminated. As the relation fails the horizontal line test as well, then it is classified as many-to-many. So (D)

(b) I. is false. Counterexample:

$$f(x) = x^2 - 2\sqrt{2}x + 2, \text{ i.e., } f(x) = (x - \sqrt{2})^2 \text{ whose zeroes are } \sqrt{2}, \sqrt{2} \text{ or equivalently } \sqrt{2}(2).$$

It is true. As α and β are zeroes of $f(x)$, then: -

(b) Circle your correct alternative.

$$a\alpha^2 + b\alpha + c = 0 \quad \& \quad 1$$

$$a\beta^2 + b\beta + c = 0$$

Consider the following two statements:

- I. Two irrational zeroes of a quadratic function cannot be equal.
- II. If α and β are distinct zeroes of the quadratic function $f(x) = ax^2 + bx + c$, where a, b and c are real constants, then $a\alpha + b + a\beta = 0$.

Which of the above statements are true?

- (A) Neither I nor II
- (B) I only
- ☒ (C) II only
- (D) Both I and II

Subtracting the above equations yields $a(\alpha^2 - \beta^2) + b(\alpha - \beta) = 0$
 $\Rightarrow a(\alpha - \beta)(\alpha + \beta) + b(\alpha - \beta) = 0$
 $\Rightarrow a(\alpha + \beta) + b = 0$ as $\alpha \neq \beta$
 $\Rightarrow a\alpha + b + a\beta = 0$, as required.
 $\therefore (C)$

Question 2 Continued

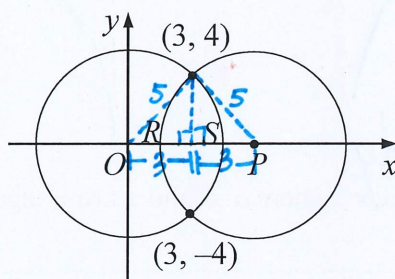
(c) Solve $\frac{x-5}{4} = \frac{6}{x}$.

2

$$\begin{aligned} \Rightarrow x(x-5) &= 24 \\ \Leftrightarrow x^2 - 5x - 24 &= 0 \quad \checkmark \text{ correct quadratic equation} \\ \Leftrightarrow (x-8)(x+3) &= 0 \\ \Leftrightarrow x &= -3 \text{ or } 8 \quad \checkmark \text{ correct answers.} \end{aligned}$$

(d)

2



Not to Scale

In the diagram above, O and P are centres of circles with equal radii, which intersect at $(3, 4)$ and $(3, -4)$. The circles cut the x axis at R and S as shown.

Find the value of the length RS . Explain your reasoning carefully.

Radius of both circles found by a simple application of Pythagoras' theorem or the distance formula applied to $O(0,0)$ & $(3,4)$. Radius = 5 units.
 From the same right-angled triangle constructions, it is evident that $OP = 3 + 3$ or 6 units. $\therefore OR = OP - RP = 6 - 5$ or 1 unit. Finally, $RS = OS - OR$
 $\checkmark$ Correct answer $= \text{Radius} - OR$
 $\checkmark$ Valid reasoning. $= 5 - 1$ or 4 units.

End of Question 2

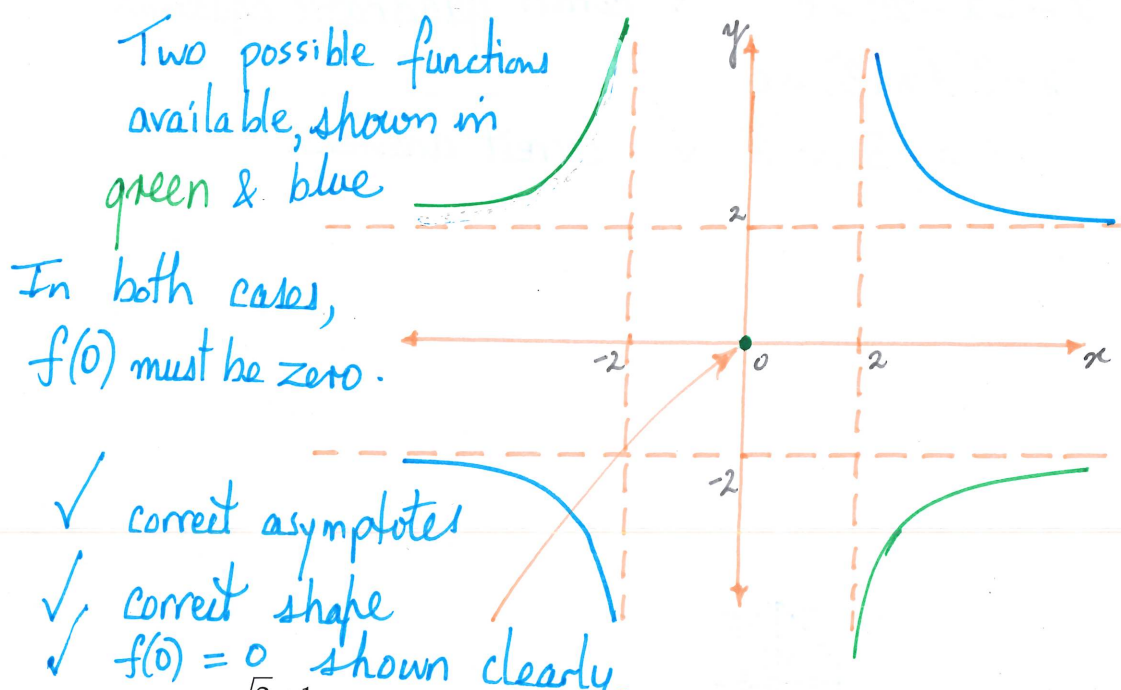
Question 3 (4 marks)

Marks

(a) Sketch a function $y = f(x)$ that possesses all of the following properties:

3

1. $f(x)$ is an odd function.
2. $f(x)$ has horizontal asymptotes $y = \pm 2$.
3. $f(x)$ has vertical asymptotes $x = \pm 2$.
4. $f(x)$ has domain $x < -2$ together with $x > 2$ as well as $x = 0$.



(b) Write $\frac{\sqrt{2}+1}{\sqrt{2}-1}$ in the form of $a + b\sqrt{x}$, where a , b and x are integers.

1

$$\begin{aligned} \frac{\sqrt{2}+1}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1} &= \frac{(\sqrt{2}+1)^2}{1} \\ &= (\sqrt{2})^2 + 2 \times 1 \times \sqrt{2} + 1^2 \\ &= \underline{\underline{3 + 2\sqrt{2}}} \quad \checkmark \text{ Correct answer.} \end{aligned}$$

End of Question 3

Question 4 (4 marks)

Marks

A function is defined by the formula $f(x) = \frac{\sqrt{-x}}{\sqrt{x+1}}$.

- (a) Determine the domain of the function.

2

Require $-x \geq 0$ & $x+1 > 0$ ✓ Correct inequalities
 $\Rightarrow x \leq 0$ & $x > -1$ } ✓ Correct simplification.
 i.e. $-1 < x \leq 0$

- (b) Determine the range of the function.

1

Clearly $f(x)$ is non-negative. Indeed, when $x=0$, $f(0)=0$ and $f \rightarrow +\infty$ as $x \rightarrow -1^+$ (Calculator check is available.)
 So range is $y \geq 0$ ✓ Correct answer

- (c) Given f is a one-to-one function, what is the relationship between x_1 and x_2 , if $f(x_1) = f(x_2)$? Explain your reasoning carefully.

1

You may assume x_1 and x_2 are in the domain of f .

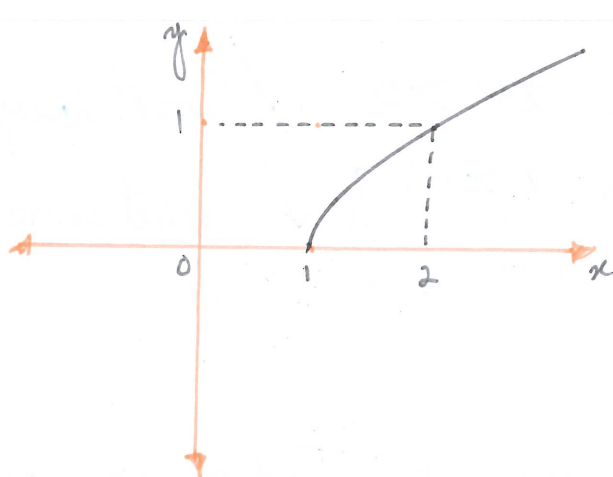
As f is a 1-1 function, then any horizontal line cuts the curve at most once. So if $f(x_1) = f(x_2)$, then $x_1 = x_2$.

✓ For $x_1 = x_2$ with valid reasoning.

End of Question 4

Question 5 (4 marks)

- (a) If $f(x) = \sqrt{x}$ sketch the graph of $y = f(x-1)$ showing intercepts with the coordinate axes. 1



✓ correct shape with
x-intercept at $x = 1$.

- (b) $P(1, 2)$ is a point on the line $3x + y - 5 = 0$. Find the coordinates of the points on this line, so that their distance from P is $\sqrt{10}$ units. 3

✓
1 mark { Let $Q(x, y)$ be any other point on the line $3x + y - 5 = 0$.
Applying the distance formula: -
$$(x-1)^2 + (y-2)^2 = (\sqrt{10})^2 \quad [PQ^2 = 10]$$

$$\Rightarrow (x-1)^2 + (y-2)^2 = 10 \quad \dots (1)$$

1 mark { However, as $Q(x, y)$ lies on the line, then $y = 5 - 3x$
(from $3x + y - 5 = 0$). So (1) becomes: -
$$(x-1)^2 + (5-3x-2)^2 = 10$$

$$\Leftrightarrow (x-1)^2 + (3-3x)^2 = 10$$

$$\Leftrightarrow x^2 - 2x + 1 + 9 - 18x + 9x^2 = 10$$

$$\Leftrightarrow 10x^2 - 20x + 10 = 10$$

$$\Leftrightarrow 10x(x-2) = 0$$

$$\Rightarrow \begin{cases} x = 0 \text{ or } x = 2 \\ y = 5 \text{ or } y = -1 \text{ respectively.} \end{cases}$$

$$\therefore \text{required co-ordinates are } (0, 5) \text{ \& } (2, -1).$$

✓ correct answers.

End of Question 5

Question 6 (4 marks)

- (a) Simplify completely $\sqrt{54x} \div \sqrt{6}$

1

$$= \frac{\sqrt{54x}}{\sqrt{6}} \text{ or } \sqrt{\frac{54x}{6}}$$

$$= \sqrt{9x} \text{ or } \underline{3\sqrt{x}} \quad \checkmark \text{ for correct answer}$$

- (b) Factorise fully $a^2 - 2ab + b^2 + 3a - 3b + 2$.

1 mark.

3

$$= (a-b)^2 + 3a - 3b + 2$$

$$= (a-b)^2 + 3(a-b) + 2$$

Let $X = a-b$. Algebra is equivalent to

$$X^2 + 3X + 2$$

$$\Rightarrow (X+2)(X+1)$$

$$\Rightarrow (a-b+2)(a-b+1)$$

2 marks
or
equivalent
logic.

End of Question 6

Question 7 (4 marks)

- (a) How many integers are between $\sqrt{50}$ and $\sqrt{500}$?

1

Noting $\sqrt{8^2} = \sqrt{64}$ or 8 is the least & $\sqrt{22^2} = \sqrt{484}$ or 22 is the largest, we require an inclusive count from 8 through to 22. This is $22 - 8 + 1$ or 15. Hence, there are 15 integers between $\sqrt{50}$ & $\sqrt{500}$.
1 mark

- (b) By definition, a positive integer that can be written as the square of a positive integer is called a square number or a perfect square. For example, 9 is a perfect square since $9 = 3^2$.

3

For x a positive integer, how many positive integers are there such that both x and $x + 99$ are perfect squares? Explain carefully your reasoning.

Hint: Let $x = m^2$ and $x + 99 = n^2$, where m and n are positive integers.

Using the hint $n^2 - m^2 = (x + 99) - x$

i.e., $(n + m)(n - m) = 99$; $n > m$

1 mark or equivalent

Consider the factors of 99 which are 1, 3, 9, 11, 33 & 99.

Thus, possible combinations for n & m are:-

$$\begin{cases} n+m=99 \\ n-m=1 \end{cases}, \begin{cases} n+m=33 \\ n-m=3 \end{cases}, \begin{cases} n+m=11 \\ n-m=9 \end{cases}$$

Solving these systems of simultaneous equations gives:-

$n=50, m=49$; $n=18, m=15$; $n=10, m=1$ respectively.

Thus, x could be $\begin{cases} 49^2 \text{ or } 2401 \\ 15^2 \text{ or } 225 \\ 1^2 \text{ or } 1 \end{cases}$

Consequently, there are exactly three positive integers satisfying the criteria.

1 mark for systems of linear equations

1 mark for correct answer/conclusion following from argument

End of Assessment Task

N.B. No credit for "correct" bald answer.